UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Applications of Linear Differential Equations: LCR Circuits

Complete study pack for class 014 (deck SESSION 10): the voltage drops $L\frac{dI}{dt}$, RI, $\frac{Q}{C}$, Kirchhoff's voltage law, the charge equation $L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t)$ with $I = \frac{dQ}{dt}$, initial conditions, damping by R^2 against $\frac{4L}{C}$, transient and steady state, and the first-order RL and RC circuits of the class-work sheet

Compiled from the PESU Unit 2 class 014 slide deck (17 slides), its AV-summary transcript (11.6 min), the Unit 1 & 2 class-work sheet (problems 34–36) and eight past ESA papers. Generated 17 October 2026.

· How to use this pack

Topic 1 builds the model: three voltage drops, Kirchhoff's voltage law, the charge equation (E1.3) and the current equation obtained by differentiating it (E1.5). Topic 2 solves it: CF from the auxiliary equation (its roots decide over-, critically or underdamped, table C2.1), PI from the forcing, then the constants from $Q(0)$ and $I(0)$ (Figure F1.1); it works both slide examples. Topic 3 solves the two ESA questions (Jan-24 sets 3 and 4) and the free discharge of CW P36. Topic 4 is the first-order RL and RC circuits of CW P34–P35. The CF rules are in the class 002 pack and the PI rules for constants and sin, cos in the class 003 and 004 packs; every worked example here is nevertheless complete. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S014·8	Unit 2 deck 014 (“Applications of DE – LCR circuit”), slide 8 (rendered .pptx)
T014 00:04:29	Unit 2 AV-summary transcript “Applications of higher-order linear differential equations – RLC Circuits” (11.6 min), segment start
CW U2 P34	Unit 2 class-work problem 34
ESA Jan-24 s3 3b	Past paper: UE23MA141A Jan 2024 set 3, question 3.b

Contents

1. 1 · Modelling the series LCR circuit
2. 2 · Solving the circuit equation: CF, PI and damping
3. 3 · Past-paper circuits: AC sources and free discharge
4. 4 · First-order circuits: RL and RC
5. Appendix A · Symbol glossary
6. Appendix B · Sources, coverage and gaps

1 Modelling the series LCR circuit

A series LCR circuit has an inductor L (henries), a resistor R (ohms) and a capacitor C (farads) driven by an electromotive force $E(t)$ (volts). The voltage drops across them are $L\frac{dI}{dt}$, RI and $\frac{Q}{C}$, and Kirchhoff's voltage law says they add up to $E(t)$. With $I = \frac{dQ}{dt}$ this becomes the second-order linear equation with constant coefficients $L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t)$; differentiating it once gives the same kind of equation for the current. Two initial conditions, $Q(0)$ and $I(0)$, fix the solution.

Key terms and definitions

Series LCR (RLC) circuit. An inductor of inductance L (henries), a capacitor of capacitance C (farads) and a resistor of resistance R (ohms) connected in series with a source of electromotive force $E(t)$ (volts). S014·4 · T014 00:00:13

Current $I(t)$. The current through the whole circuit, in amperes; $I = \frac{dQ}{dt}$. S014·4 · S014·6 · T014 00:02:38

Charge $Q(t)$. The charge on the capacitor, in coulombs. S014·4 · S014·6

Voltage drop. The potential difference across one component: $L\frac{dI}{dt}$ (inductor), RI (resistor, Ohm's law), $\frac{Q}{C}$ (capacitor). S014·5 · T014 00:01:18

Kirchhoff's voltage law. The voltage supplied by the battery equals the sum of the voltage drops across the circuit components. S014·5 · T014 00:01:51

Initial conditions. The charge $Q(0)$ and the current $I(0) = Q'(0)$ at $t = 0$; they fix c_1 and c_2. S014·7 · T014 00:03:41

> *"A simple RLC circuit consists of an inductor of inductance 'L' (in henries), a Capacitor of capacitance 'C' (in farads) and a resistor of resistance 'R' (in ohms) and a source of electromotive force E(t) (in volts) are connected in series. The current 'I' through the circuit is measured in amperes and the charge 'Q' on the capacitor is measured in coulombs."* S014·4

> *"From Kirchoff's Voltage Law, The voltage supplied by the battery = The voltage drop across the circuit components (the sum of these voltage drops is equal to the electromotive force)."* S014·5

> *"$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t)$. This is the differential equation which describes LCR – Circuit."* S014·6

> *"It is a second order linear differential equation with constant coefficients. If the charge and the current are known at zero time, we can get an expression for Q(t) and I(t)."* S014·7

Formulas

E1.1 · Voltage drops ESA ★★

$$E_L = L\frac{dI}{dt}, \qquad E_R = RI, \qquad E_C = \frac{Q}{C}$$

GIVES The voltage across the inductor, resistor and capacitor.

APPLIES Any circuit with these elements.

UNITS volts; L in H, R in Ω, C in F, I in A, Q in C

NOTE $E_R = RI$ is Ohm's law.

S014·5 · T014 00:01:18 · T014 00:01:34

E1.2 · Kirchhoff's voltage law ESA ★★

$$E(t) = V_L + V_R + V_C = L\frac{dI}{dt} + RI + \frac{Q}{C}$$

GIVES The circuit equation in I and Q.

APPLIES Series circuit.

NOTE Sum of the drops = the EMF supplied.

S014·5 · S014·6 · T014 00:01:51 · T014 00:02:21

E1.3 · The LCR equation for the charge ESA ★★

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t), \qquad I = \frac{dQ}{dt}$$

GIVES A second-order linear DE with constant coefficients for $Q(t)$.

APPLIES Series LCR circuit; L, R, C constant.

NOTE Divide by L before finding the AE: $Q'' + \frac{R}{L}Q' + \frac{1}{LC}Q = \frac{E}{L}$.

S014·6 · S014·7 · S014·14 · T014 00:02:58 · ESA Jan-24 s3 3b · ESA Jan-24 s4 3b

E1.4 · Initial conditions ESA ★★

$$Q(0) = Q_0, \qquad I(0) = Q'(0) = I_0$$

GIVES The two conditions that fix c_1, c_2.

APPLIES Usually $Q_0 = I_0 = 0$ (switch closed at $t = 0$).

NOTE Use $Q(0)$ in $Q(t)$, then differentiate $Q(t)$ and use $I(0)$.

S014·7 · S014·11 · S014·12 · T014 00:03:41 · T014 00:08:21

E1.5 · The equation for the current ESA ★

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{I}{C} = \frac{dE}{dt}, \qquad LI'(0) = E(0) - RI(0) - \frac{Q(0)}{C}$$

GIVES $I(t)$ directly, by differentiating E1.3 once.

APPLIES The current is asked and E is differentiable.

NOTE The second condition comes from E1.2 at $t = 0$. A constant E gives $\frac{dE}{dt} = 0$.

ADDED FOR CLARITY S014·5 · S014·16

Graphs and diagrams

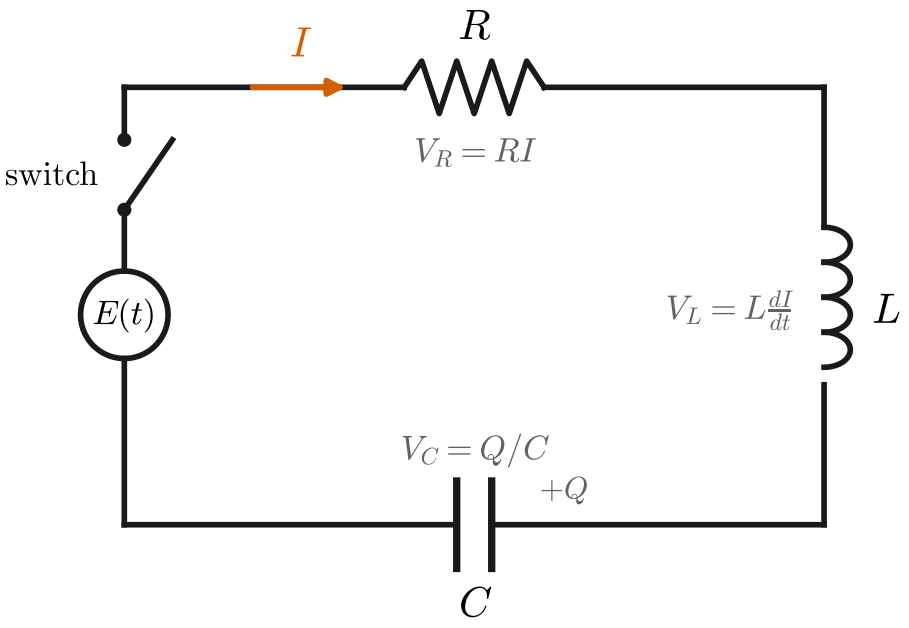
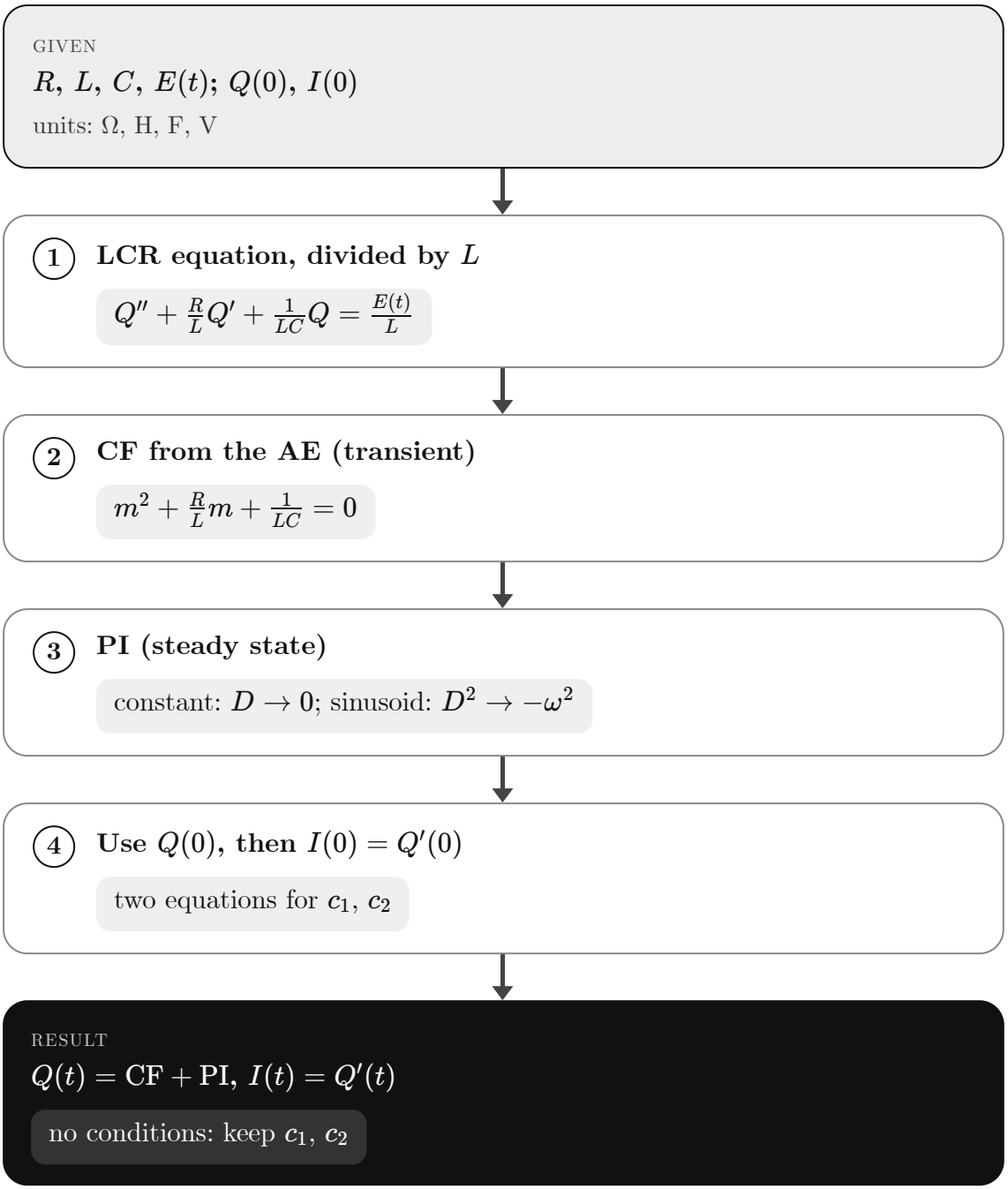


Figure G1.1. The series LCR circuit (slide 014·4, redrawn). ESA ★★

S014·4 · T014 00:00:13

How to read it. Source $E(t)$ and switch on the left, resistor R on top, inductor L on the right, capacitor C at the bottom, all in one loop so the same current I flows through each. The three drops $V_R = RI$, $V_L = L\frac{dI}{dt}$, $V_C = \frac{Q}{C}$ add up to $E(t)$. Schematic, not to scale.

Flowcharts



Remark. Steps 2 and 3 are the class 002 and 003–004 methods.

Exam trap. Divide $E(t)$ by L too.

Figure F1.1. From circuit data to $Q(t)$ and $I(t)$ ESA ★★

S014·8 · S014·9 · S014·10 · S014·11 · S014·12 · T014 00:04:57 · T014 00:10:31

The three circuit elements

Element	Symbol, unit	Voltage drop	In terms of Q
Inductor	L, henry (H)	$L\frac{dI}{dt}$	$L\frac{d^2Q}{dt^2}$
Resistor	R, ohm (Ω)	RI	$R\frac{dQ}{dt}$
Capacitor	C, farad (F)	$\frac{Q}{C}$	$\frac{Q}{C}$
Source	$E(t)$, volt (V)	supplies $E(t)$	right-hand side

S014·4 · S014·5

Important points and why

1. Three drops, one law: $L\frac{dI}{dt} + RI + \frac{Q}{C} = E(t)$; then substitute $I = \frac{dQ}{dt}$ to get one equation in Q. ESA ★★ S014·5 · S014·6 · T014 00:02:38
2. The LCR equation is a second-order non-homogeneous linear DE with constant coefficients: solve it with CF + PI. ESA ★★ S014·7 · T014 00:03:21 · T014 00:06:03
3. Two constants need two conditions: the charge and the current at $t = 0$. ESA ★★ S014·7 · T014 00:03:41
4. Substitute the numbers first and divide by L: $2Q'' + 8Q' + \frac{Q}{0.1} = 10 \Rightarrow Q'' + 4Q' + 5Q = 5$. ESA ★★ S014·8 · S014·9 · T014 00:05:46
5. Watch $\frac{Q}{C}$: $\frac{1}{0.1} = 10$, $\frac{1}{16\times10^{-4}} = 625$, $\frac{1}{0.002} = 500$, $\frac{1}{10^{-3}} = 1000$. ESA ★★ S014·9 · ESA Jan-24 s3 3b · ESA Jan-24 s4 3b
6. To get the current, differentiate the final $Q(t)$; or solve E1.5 for I directly (W1.1). ESA ★★ S014·12 · T014 00:10:31

Why does $I = \frac{dQ}{dt}$ turn E1.2 into a second-order equation?
$L\frac{dI}{dt}$ becomes $L\frac{d^2Q}{dt^2}$, RI becomes $R\frac{dQ}{dt}$, and $\frac{Q}{C}$ stays: one unknown, Q, with derivatives up to order 2.
S014·6 · T014 00:02:58

Exam trap. $\frac{Q}{0.1} = 10Q$, not $0.1Q$; and $\frac{1}{16\times10^{-4}} = 625$, not $\frac{1}{16}$. S014·9 · ESA Jan-24 s3 3b

Exam trap. Divide the whole equation by L, including $E(t)$: $Q'' + 4Q' + 5Q = 5$ (not 10). S014·9 · T014 00:05:46

Worked examples

W1.1 · The current equation for the slide example ESA ★

GIVEN $R = 8\ \Omega$, $L = 2$ H, $C = 0.1$ F, $E = 10$ V, $Q(0) = I(0) = 0$. Find $I(t)$ without finding Q first.

STEP 1 Differentiate $2Q'' + 8Q' + 10Q = 10$: $2I'' + 8I' + 10I = 0$, i.e. $I'' + 4I' + 5I = 0$.

STEP 2 Second condition from E1.2 at $t = 0$: $2I'(0) = 10 - 8 \cdot 0 - 10 \cdot 0$, so $I'(0) = 5$.

STEP 3 AE $m^2 + 4m + 5 = 0$, $m = -2 \pm i$: $I = e^{-2t}(a\cos t + b\sin t)$.

STEP 4 $I(0) = a = 0$; $I'(0) = -2a + b = 5$, so $b = 5$.

ANSWER $\boxed{I(t) = 5e^{-2t}\sin t}$

CHECK Same as slide 014·12 (sympy).

ADDED FOR CLARITY S014·8 · S014·12

SHORT-NOTE RECAP

Series LCR circuit: drops $L\frac{dI}{dt}$, RI, $\frac{Q}{C}$; Kirchhoff's voltage law: their sum is $E(t)$. With $I = \frac{dQ}{dt}$: $LQ'' + RQ' + \frac{Q}{C} = E(t)$, a second-order linear DE with constant coefficients. Units: H, Ω, F, V, A, C. Conditions $Q(0)$, $I(0) = Q'(0)$. Differentiating gives $LI'' + RI' + \frac{I}{C} = E'(t)$ with $LI'(0) = E(0) - RI(0) - \frac{Q(0)}{C}$.

2 Solving the circuit equation: CF, PI and damping

The charge is $Q = \text{CF} + \text{PI}$. The CF comes from the auxiliary equation $Lm^2 + Rm + \frac{1}{C} = 0$; because $L, R, C > 0$ it always decays, so it is the transient. Its roots are real and distinct, equal, or complex according as R^2 is greater than, equal to or less than $\frac{4L}{C}$ (over-, critically or underdamped). The PI is the steady state that remains: a constant CE for a battery, a sinusoid of the source frequency for an AC source. The slides work a battery example ($Q = 1 + e^{-2t}(-\cos t - 2\sin t)$) and an AC example.

Key terms and definitions

Transient. The CF part of $Q(t)$; it decays to zero because every root has a negative real part. ADDED FOR CLARITY S014·10 · S014·16

Steady state. The PI part, which remains for large t: CE for a constant source, a sinusoid at the source frequency for an AC source. ADDED FOR CLARITY S014·10 · S014·15

Overdamped, critically damped, underdamped. $R^2 > \frac{4L}{C}$ (two real roots, no oscillation), $R^2 = \frac{4L}{C}$ (equal roots), $R^2 < \frac{4L}{C}$ (complex roots, decaying oscillation). ADDED FOR CLARITY S014·9 · S014·15

Formulas

E2.1 · Auxiliary equation and its roots ESA ★★

$$Lm^2 + Rm + \frac{1}{C} = 0, \qquad m = \frac{-R \pm \sqrt{R^2 - 4L/C}}{2L}$$

GIVES The CF (transient) of the charge.
APPLIES Every LCR problem.
NOTE Slide example: $2m^2 + 8m + 10 = 0$, $m = -2 \pm i$.
S014·9 · S014·15 · T014 00:06:48

E2.2 · Damping regimes ESA ★

$$R^2 > \tfrac{4L}{C}:\ Q_{tr} = c_1 e^{m_1 t} + c_2 e^{m_2 t}$$
$$R^2 = \tfrac{4L}{C}:\ Q_{tr} = (c_1 + c_2 t) e^{-Rt/(2L)}$$
$$R^2 < \tfrac{4L}{C}:\ Q_{tr} = e^{-Rt/(2L)}(c_1 \cos \omega_d t + c_2 \sin \omega_d t)$$

GIVES The form of the transient.
APPLIES Compare R^2 with $\frac{4L}{C}$.
NOTE Over-, critically, underdamped. All decay: $m_1, m_2 < 0$ or real part $-\frac{R}{2L} < 0$.
ADDED FOR CLARITY S014·9 · S014·15

E2.3 · Damped frequency ESA ★

$$\omega_d = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}, \qquad \omega_0 = \frac{1}{\sqrt{LC}}\ (R = 0)$$

GIVES The angular frequency of the underdamped transient.

APPLIES $R^2 < \frac{4L}{C}$.

UNITS rad/s

NOTE ESA s3: $\sqrt{625 - 400} = 15$; s4: $\sqrt{500 - 100} = 20$.

ADDED FOR CLARITY S014·9 · ESA Jan-24 s3 3b

E2.4 · Solution = transient + steady state ESA ★★

$$Q(t) = \underbrace{\text{CF}}_{\text{transient}} + \underbrace{\text{PI}}_{\text{steady state}}, \qquad E = \text{const} \Rightarrow \text{PI} = CE$$

GIVES The structure of every answer.

APPLIES Any $E(t)$.

NOTE Constant E: $\frac{E/L}{D^2 + \frac{R}{L}D + \frac{1}{LC}}$ with $D \to 0$ gives CE (slide: $\frac{5}{5} = 1 = 0.1 \times 10$).

S014·10 · T014 00:07:37 · T014 00:07:52

E2.5 · Steady-state amplitude (impedance) ESA ★

$$E = E_0 \sin \omega t: \quad I_{ss} = \frac{E_0}{Z}\sin(\omega t - \theta), \quad Z = \sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}, \quad |Q_{ss}| = \frac{E_0}{\omega Z}$$

GIVES A check on any PI.

APPLIES Sinusoidal source.

NOTE ESA s3: $Z = \sqrt{1600 + 120^2} = 126.5$, $|Q_{ss}| = \frac{100}{5 \cdot 126.5} = 0.158 = \sqrt{0.15^2 + 0.05^2}$.

ADDED FOR CLARITY ESA Jan-24 s3 3b · S014·15

Graphs and diagrams

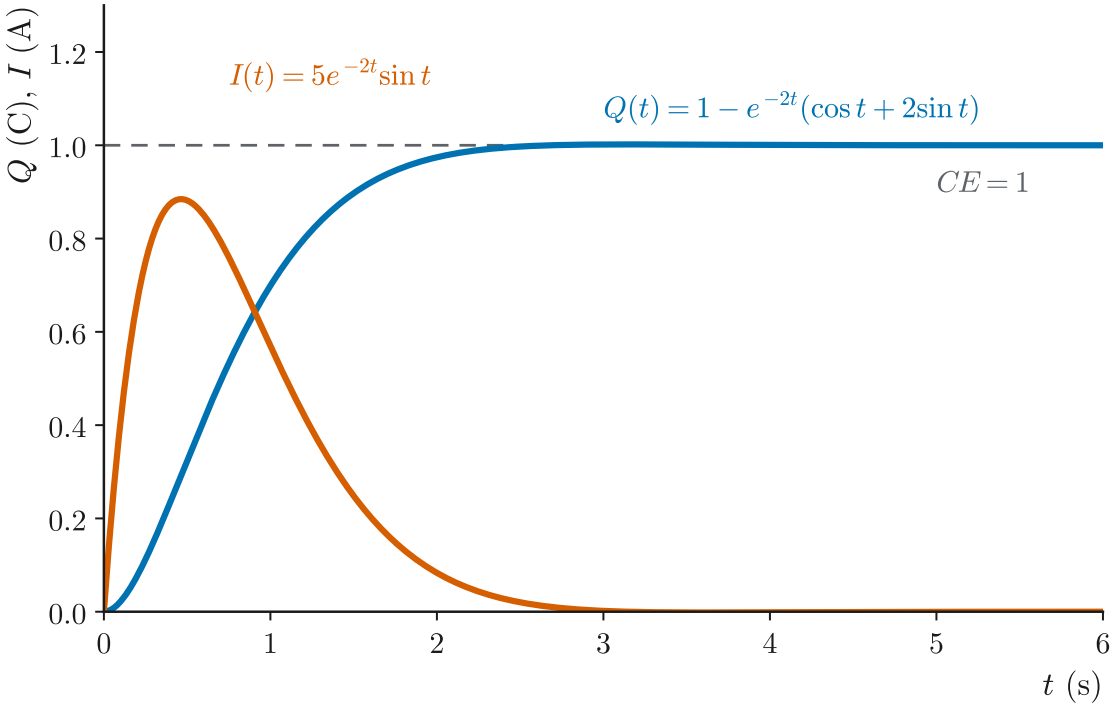


Figure G2.1. Slide example: $Q(t)$ and $I(t)$ for $R = 8$, $L = 2$, $C = 0.1$, $E = 10$. ESA ★★

ADDED FOR CLARITY S014·12

How to read it. The charge (blue) rises from 0 to the steady state $CE = 1$ C (grey dashed) with a small overshoot (underdamped, e^{-2t}). The current $5e^{-2t}\sin t$ (vermilion) starts at 0, peaks near $t = 0.46$ s and dies out: a battery drives no steady current through a capacitor.

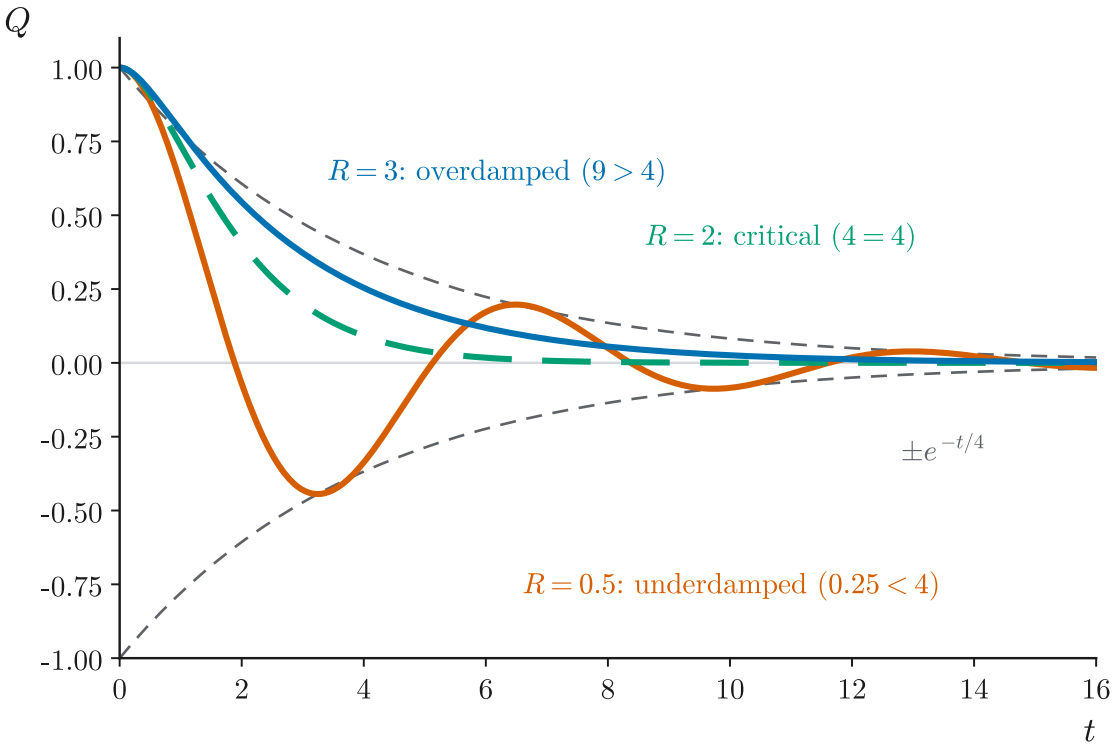


Figure G2.2. Damping regimes: R^2 against $\frac{4L}{C}$. ESA ★

ADDED FOR CLARITY S014·9 · S014·15

How to read it. Free discharge $Q'' + RQ' + Q = 0$ ($L = C = 1$, so $\frac{4L}{C} = 4$) from $Q(0) = 1$, $I(0) = 0$. $R = 3$ (blue, overdamped) creeps back slowly; $R = 2$ (green dashed, critical) returns fastest without crossing zero; $R = 0.5$ (vermilion, underdamped) oscillates inside the envelope $\pm e^{-t/4}$ (grey).

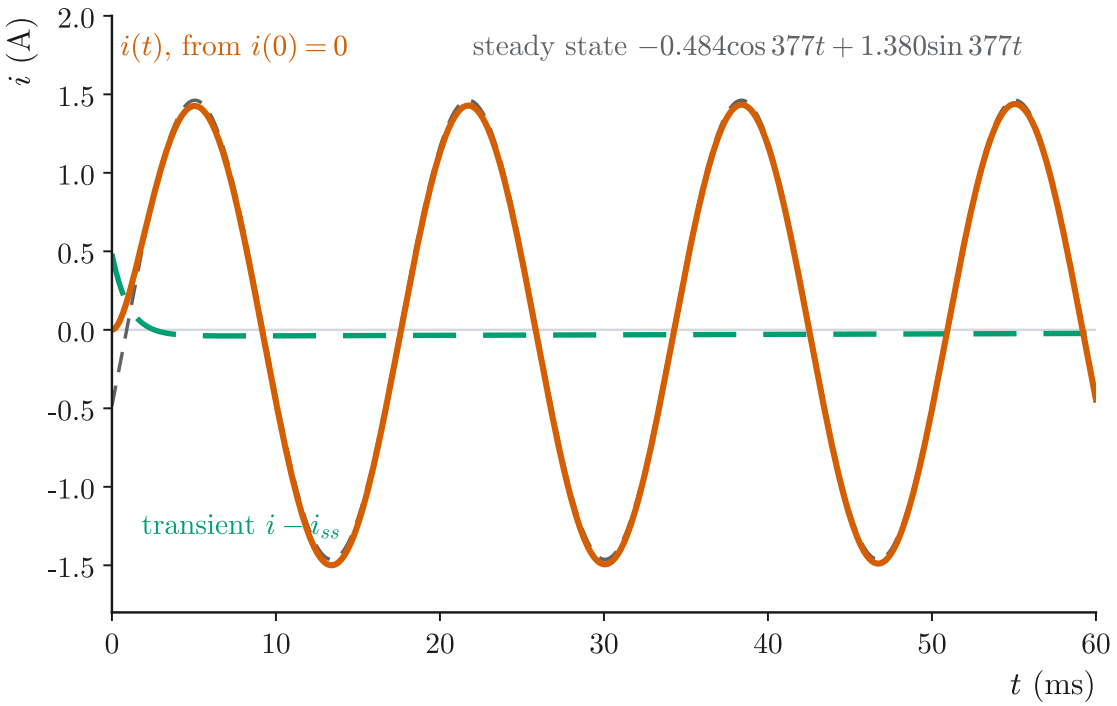
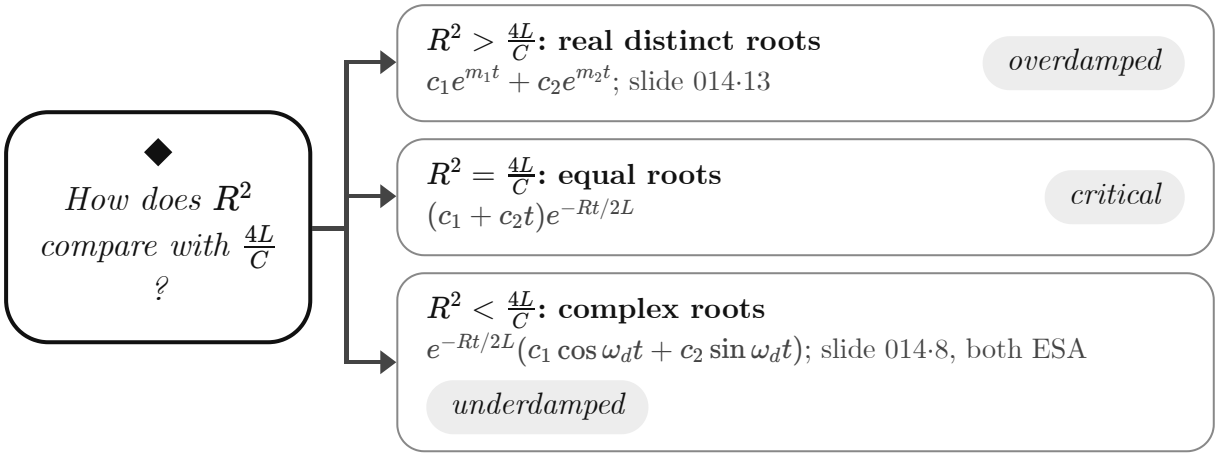


Figure G2.3. Slide 014·16: current for $E = 155\sin 377t$, transient and steady state. ESA ★

ADDED FOR CLARITY S014·16

How to read it. The full current $i(t)$ (vermilion), the steady state $-0.4838\cos 377t + 1.3804\sin 377t$ (grey dashed) and their difference, the transient (green dashed). The fast part ($e^{-989.9t}$) is gone within a few milliseconds; the slow one ($-0.0424e^{-10.1t}$) shifts the curve slightly for a few tenths of a second.

Flowcharts



Remark. All three decay; only the underdamped one oscillates.

Figure F2.1. Which transient? Compare R^2 with $\frac{4L}{C}$ ESA ★

ADDED FOR CLARITY S014·9 · S014·15

Damping regimes

Test	Roots	Transient	Example
$R^2 > \frac{4L}{C}$	real, distinct, negative	$c_1 e^{m_1 t} + c_2 e^{m_2 t}$	slide 014·13: $10^4 > 400$
$R^2 = \frac{4L}{C}$	equal, $-\frac{R}{2L}$	$(c_1 + c_2 t)e^{-Rt/2L}$	$R = 2$, $L = C = 1$ (G2.2)
$R^2 < \frac{4L}{C}$	$-\frac{R}{2L} \pm i\omega_d$	$e^{-Rt/2L}(c_1 \cos \omega_d t + c_2 \sin \omega_d t)$	slide 014·8; ESA s3, s4; CW P36

ADDED FOR CLARITY S014·9 · S014·15 · ESA Jan-24 s3 3b

Every circuit problem of this class, side by side

Problem	Normalised equation	Roots	PI (steady state)
Slide 014·8	$Q'' + 4Q' + 5Q = 5$	$-2 \pm i$	1
Slide 014·13	$\ddot{q} + 1000\dot{q} + 10^4 q = 1550 \sin 377t$	-10.10, -989.9	$-0.00128 \sin 377t - 0.00366 \cos 377t$
ESA s3 3b	$Q'' + 40Q' + 625Q = 100 \cos 5t$	$-20 \pm 15i$	$0.15 \cos 5t + 0.05 \sin 5t$
ESA s4 3b	$Q'' + 20Q' + 500Q = 12 \sin 10t$	$-10 \pm 20i$	$0.024 \sin 10t - 0.012 \cos 10t$
CW P36	$q'' + 1000q' + 2 \times 10^6 q = 0$	$-500 \pm 500\sqrt{7}i$	0 (free)

S014·8 · S014·13 · ESA Jan-24 s3 3b · ESA Jan-24 s4 3b · CW U2 P36

Important points and why

1. $Q(0)$ gives c_1 at once; for c_2 differentiate $Q(t)$ (product rule on $e^{-2t}(\dots)$) and use $I(0)$. ESA ★★ S014·11 · S014·12 · T014 00:09:00
2. For a battery (E constant) the PI is a constant: replace D by 0, giving $Q_{ss} = CE$, and $I \to 0$. ESA ★★ S014·10 · T014 00:07:52
3. R^2 against $\frac{4L}{C}$ decides the CF: real distinct (overdamped), equal (critical), complex (underdamped). Slide 014·8: $64 < 80$, underdamped; slide 014·13: $10^4 > 400$, overdamped. ESA ★ ADDED FOR CLARITY S014·9 · S014·15
4. The CF always decays (all roots have negative real part when $R > 0$), so it is the transient; the PI is what you measure after a while. ESA ★★ ADDED FOR CLARITY S014·16
5. With an AC source the steady state has the source frequency, not ω_d; its amplitude is largest near $\omega = \frac{1}{\sqrt{LC}}$ (resonance). ESA ★ ADDED FOR CLARITY S014·15

Why does the transient always die out?
The roots of $Lm^2 + Rm + \frac{1}{C} = 0$ have sum $-\frac{R}{L} < 0$ and product $\frac{1}{LC} > 0$: both real roots are negative, or the complex pair has real part $-\frac{R}{2L} < 0$. So every CF term carries a decaying exponential.

ADDED FOR CLARITY S014·9 · S014·15

Why is the battery steady state $Q = CE$?
When nothing changes, $I = 0$ and $\frac{dI}{dt} = 0$, so E1.2 leaves $\frac{Q}{C} = E$: the capacitor is fully charged and blocks the current.

ADDED FOR CLARITY S014·10 · S014·12

Exam trap. $I(0) = 0$ is a condition on $Q'(0)$: differentiate $Q(t)$ (including the $-2e^{-2t}$ term of the product rule) before substituting $t = 0$. S014·11 · S014·12 · T014 00:09:19

Exam trap. Slide 014·11 writes $e^{-2.0}$: it means $e^{-2\cdot 0} = 1$, not e^{-2}. ADDED FOR CLARITY S014·11

Exam trap. Slide 014·16 uses $i(0) = 0$ and $i'(0) = 0$ but states only the first; $i'(0) = \ddot{q}(0) = 0$ comes from the equation at $t = 0$. ADDED FOR CLARITY S014·16

Worked examples

W2.1 · Battery: $R = 8$, $L = 2$, $C = 0.1$, $E = 10$ V (slides 014·8–12) ESA ★★

GIVEN Find an expression for $I(t)$ in an RLC circuit with $R = 8\ \Omega$, $L = 2$ H, $C = 0.1$ F, $E(t) = 10$ V, assuming zero charge and zero current at $t = 0$.

STEP 1 $2Q'' + 8Q' + \frac{Q}{0.1} = 10 \Rightarrow Q'' + 4Q' + 5Q = 5$.

STEP 2 AE $m^2 + 4m + 5 = 0$: $m = -2 \pm i$. CF $= e^{-2t}(c_1 \cos t + c_2 \sin t)$.

STEP 3 PI $= \frac{5e^{0t}}{D^2+4D+5} = \frac{5}{5} = 1$ $(D \to 0)$. $Q = e^{-2t}(c_1 \cos t + c_2 \sin t) + 1$.

STEP 4 $Q(0) = 0$: $c_1 + 1 = 0$, $c_1 = -1$.

STEP 5 $Q' = e^{-2t}(-c_1 \sin t + c_2 \cos t) - 2e^{-2t}(c_1 \cos t + c_2 \sin t)$; $I(0) = c_2 - 2c_1 = 0$, $c_2 = -2$.

STEP 6 $I = Q' = e^{-2t}(\sin t - 2\cos t) - 2e^{-2t}(-\cos t - 2\sin t) = 5e^{-2t}\sin t$.

ANSWER $\boxed{Q(t) = e^{-2t}(-\cos t - 2\sin t) + 1, \qquad I(t) = 5e^{-2t}\sin t}$

CHECK sympy dsolve and checkodesol agree with the slide. $R^2 = 64 < \frac{4L}{C} = 80$: underdamped; $Q \to CE = 1$ C. Figure G2.1.

S014·8 · S014·9 · S014·10 · S014·11 · S014·12 · T014 00:04:29 · T014 00:10:49

W2.2 · AC: $E = 155\sin 377t$, $R = 100$, $L = 0.1$, $C = 10^{-3}$ (slides 014·13–16) ESA ★★

GIVEN Assuming $q = 0$ and $i = 0$ at $t = 0$, in an RLC circuit with $E(t) = 155\sin(377t)$, $R = 100\ \Omega$, $L = 0.1$ H, $C = 10^{-3}$ F, determine the current $i(t)$.

STEP 1 $0.1\ddot{q} + 100\dot{q} + 1000q = 155\sin 377t \Rightarrow \ddot{q} + 1000\dot{q} + 10000q = 1550\sin 377t$.

STEP 2 AE $r^2 + 1000r + 10000 = 0$: $r = -500 \pm 200\sqrt{6} \approx -10.1021,\ -989.898$ (overdamped: $10^4 > 400$).

STEP 3 $q_p = A\sin 377t + B\cos 377t$; with $377^2 = 142129$: $-132129A - 377000B = 1550$, $377000A - 132129B = 0$, so $A = -0.00128331$, $B = -0.00366164$.

STEP 4 $i_p = \dot{q}_p = 377A\cos 377t - 377B\sin 377t = -0.483808\cos 377t + 1.38044\sin 377t$.

STEP 5 $i = C_1 e^{-10.1021t} + C_2 e^{-989.898t} + i_p$. Conditions: $i(0) = 0$ and $i'(0) = \ddot{q}(0) = 1550\sin 0 - 1000\cdot 0 - 10000\cdot 0 = 0$.

STEP 6 $C_1 + C_2 = 0.483808$, $-10.1021C_1 - 989.898C_2 + 520.426 = 0$: $C_1 = -0.0423597$, $C_2 = 0.526168$.

ANSWER $i(t) = -0.0423597e^{-10.1021t} + 0.526168e^{-989.898t} - 0.483808\cos 377t + 1.38044\sin 377t$

CHECK sympy dsolve with $q(0) = \dot{q}(0) = 0$ reproduces every slide value. The slide does not state $i'(0) = 0$. Steady amplitude $\sqrt{0.4838^2 + 1.3804^2} = 1.463 = \frac{155}{Z}$, $Z = 105.96\ \Omega$ (E2.5). Figure G2.3.

S014·13 · S014·14 · S014·15 · S014·16

SHORT-NOTE RECAP

Divide by L; AE $Lm^2 + Rm + \frac{1}{C} = 0$. $R^2 > \frac{4L}{C}$ overdamped, $=$ critical, $<$ underdamped ($e^{-Rt/2L}(c_1\cos\omega_d t + c_2\sin\omega_d t)$). CF = transient, PI = steady state; battery: PI $= CE$. Slide example ($R = 8$, $L = 2$, $C = 0.1$, $E = 10$): $Q = 1 - e^{-2t}(\cos t + 2\sin t)$, $I = 5e^{-2t}\sin t$.

3 Past-paper circuits: AC sources and free discharge

Both ESA questions give R, L, C and a sinusoidal $E(t)$ and ask for the charge at any time t: divide by L, find the complex CF roots, find the PI by the Type 2 rule ($D^2 \to -a^2$, then rationalise), and write $Q = \text{CF} + \text{PI}$ with two arbitrary constants. If $Q(0)$ and $I(0)$ are given, fix the constants. CW P36 is the free discharge of a charged capacitor ($E = 0$): only the CF, a decaying oscillation.

Key terms and definitions

Free (natural) discharge. $E(t) = 0$: a charged capacitor discharges through L and R; the charge is the CF alone. CW U2 P36

Formulas

E3.1 · PI of a sinusoidal source (Type 2) ESA ★★

$$\frac{1}{D^2 + \frac{R}{L}D + \frac{1}{LC}}\cos\omega t = \frac{\cos\omega t}{\left(\frac{1}{LC} - \omega^2\right) + \frac{R}{L}D} \quad \text{then rationalise}$$

gives	The steady-state charge.
applies	$E = E_0 \sin\omega t$ or $E_0\cos\omega t$; replace D^2 by $-\omega^2$.
note	No failure while $R > 0$ (the CF decays, the forcing does not).

ESA Jan-24 s3 3b · ESA Jan-24 s4 3b · S014·15

Graphs and diagrams

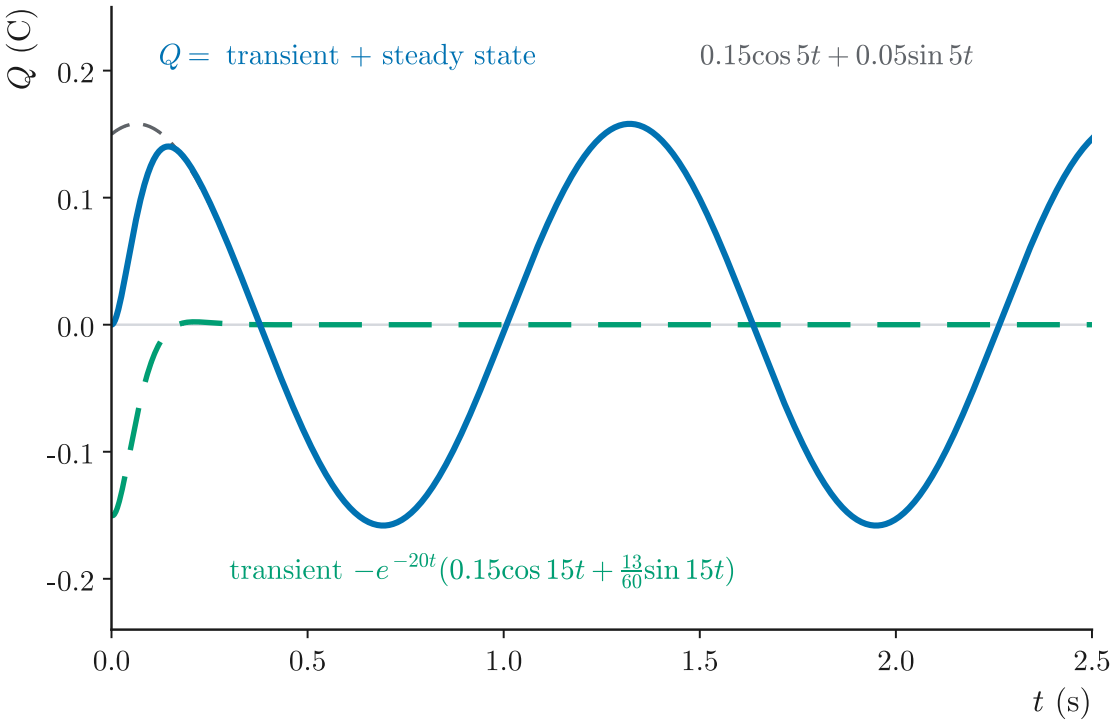


Figure G3.1. ESA $E = 100\cos 5t$ with $Q(0) = I(0) = 0$: charge = transient + steady state. ESA ★★

ADDED FOR CLARITY ESA Jan-24 s3 3b

How to read it. The charge $Q(t)$ (blue) is the sum of the transient $-e^{-20t}(0.15\cos 15t + \frac{13}{60}\sin 15t)$ (green dashed), which is gone by $t \approx 0.25$ s, and the steady state $0.15\cos 5t + 0.05\sin 5t$ (grey dashed, amplitude 0.158 C, period $\frac{2\pi}{5}$).

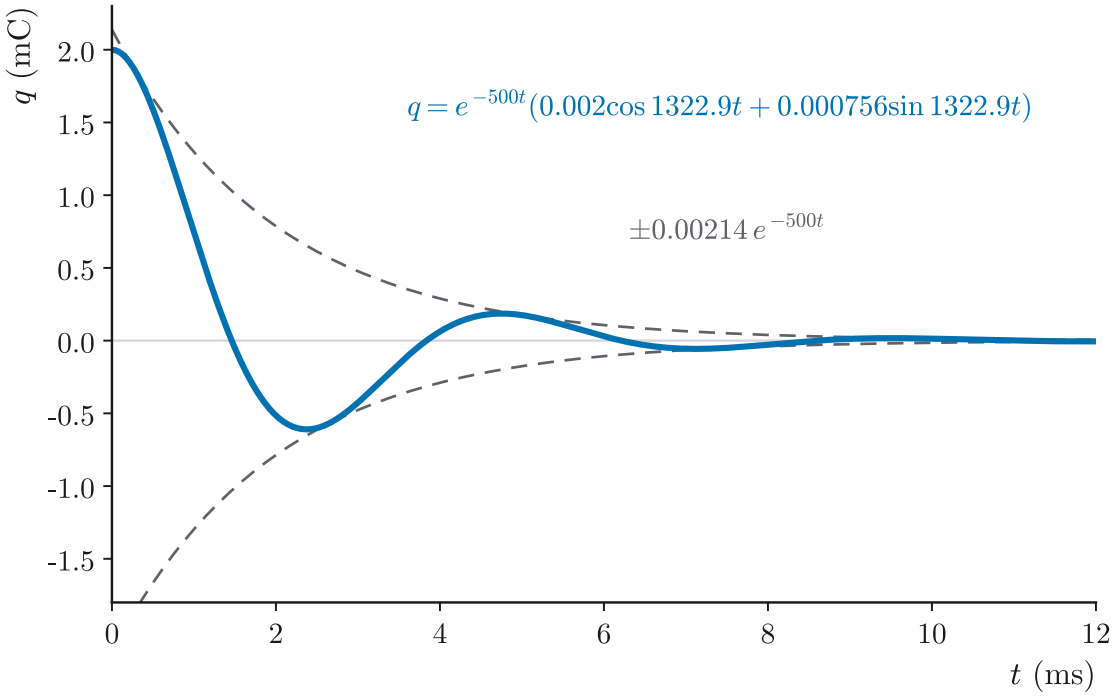
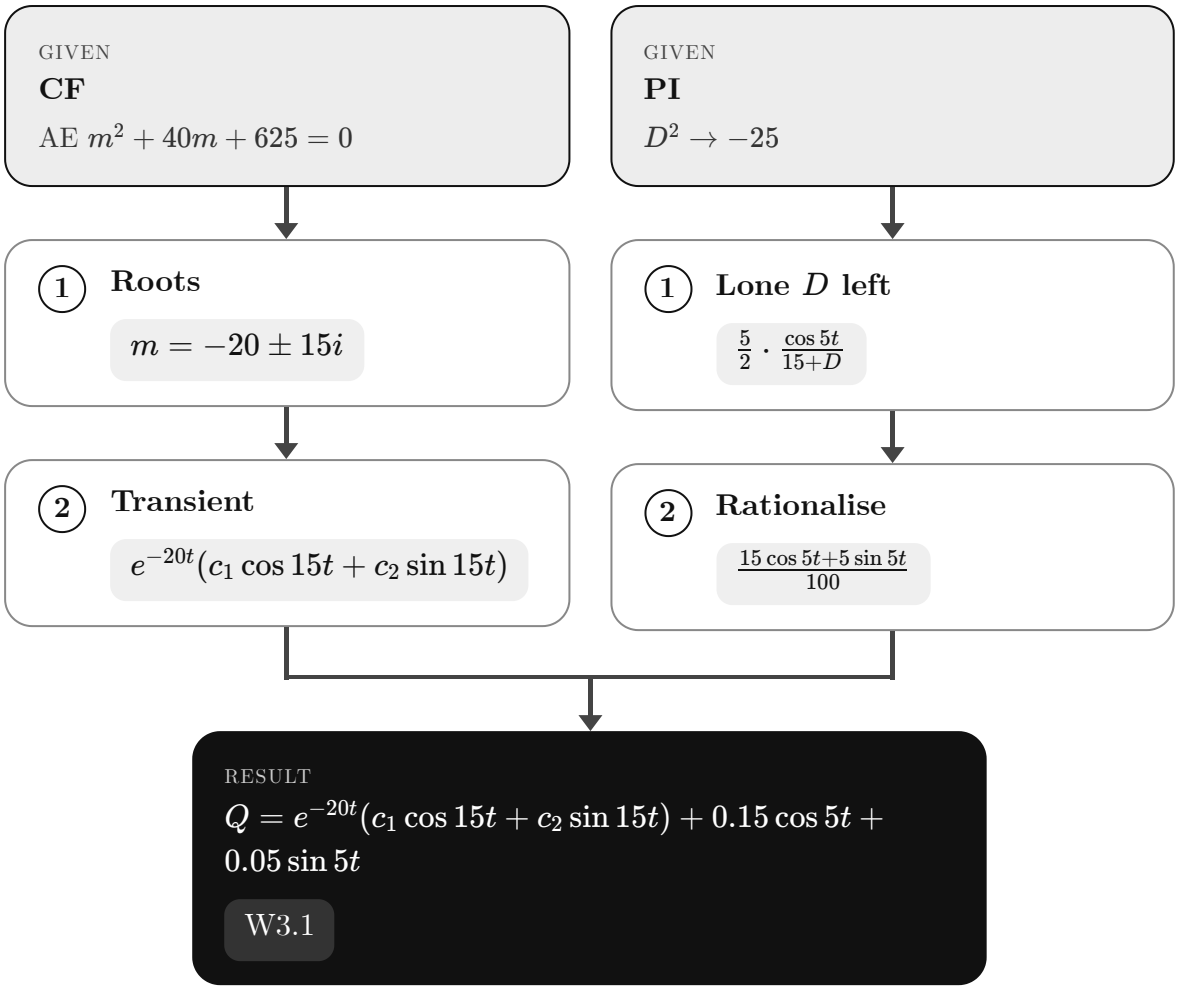


Figure G3.2. CW P36: free discharge of the capacitor. ESA ★★

ADDED FOR CLARITY CW U2 P36

How to read it. $q(t) = e^{-500t}(0.002\cos 1322.9t + 0.000756\sin 1322.9t)$ (blue) oscillates with period $\frac{2\pi}{1322.9} = 4.75$ ms inside the envelope $\pm 0.00214 e^{-500t}$ (grey dashed) and is negligible after about 10 ms.

Flowcharts



Remark. ESA s4 is the same pattern: $m = -10 \pm 20i$, PI $= 0.024 \sin 10t - 0.012 \cos 10t$ (W3.3).

Figure F3.1. ESA: $Q'' + 40Q' + 625Q = 100\cos 5t$ ESA ★★

ESA Jan-24 s3 3b

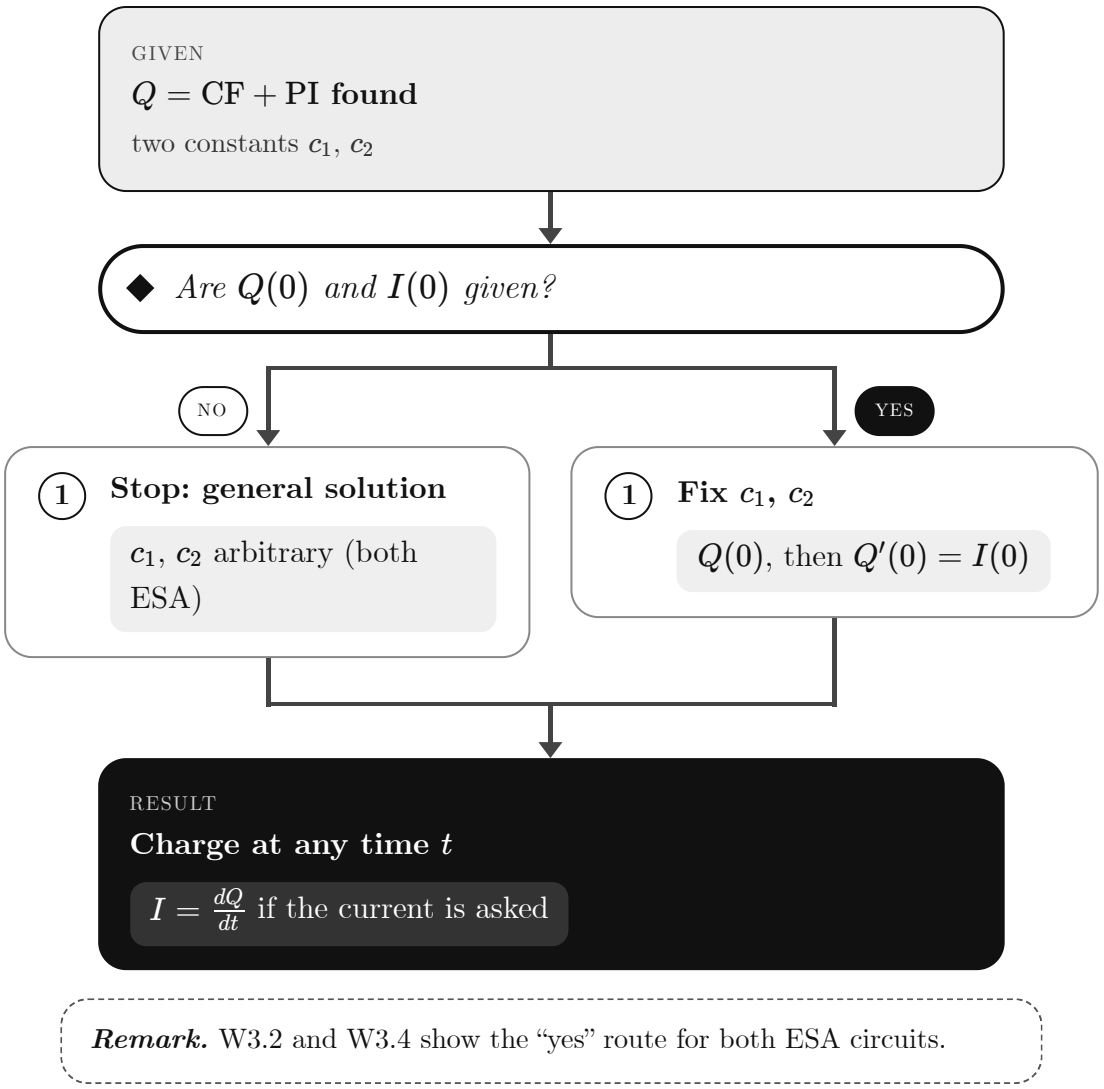


Figure F3.2. Are initial conditions given? ESA ★★

ESA Jan-24 s3 3b · ESA Jan-24 s4 3b · S014·11

Important points and why

1. ESA "find the charge at any time t" with no initial conditions: give $Q = \text{CF} + \text{PI}$ with c_1, c_2 arbitrary. ESA ★★ ESA Jan-24 s3 3b · ESA Jan-24 s4 3b
2. Both ESA circuits are underdamped: $m = -20 \pm 15i$ and $m = -10 \pm 20i$ (same as the class 002 pack). ESA ★★ ESA Jan-24 s3 3b · ESA Jan-24 s4 3b
3. PI of a sinusoid: $D^2 \to -\omega^2$, a lone D remains, rationalise with the conjugate (class 004). ESA ★★ ESA Jan-24 s3 3b · ESA Jan-24 s4 3b
4. Free discharge ($E = 0$, CW P36): $Q(0)$ is the initial charge (here 0.002 C), not 0. ESA ★★ CW U2 P36

Why can the Type 2 rule not fail in an LCR circuit with $R > 0$?
Failure needs $\pm i\omega$ to be AE roots, i.e. purely imaginary roots. With $R > 0$ the roots have negative real part, so $f(-\omega^2) \neq 0$.

ADDED FOR CLARITY ESA Jan-24 s3 3b

Exam trap. With no initial conditions do not set $c_1 = c_2 = 0$: the answer keeps the transient with arbitrary constants. ADDED FOR CLARITY ESA Jan-24 s3 3b · ESA Jan-24 s4 3b

Exam trap. CW P36's 0.0008 is $\frac{0.002}{\sqrt{7}} = 0.000756$ rounded; 1323 is $500\sqrt{7} = 1322.9$.
ADDED FOR CLARITY CW U2 P36

Worked examples

W3.1 · ESA: $R = 40$, $L = 1$, $C = 16 \times 10^{-4}$, $E = 100\cos 5t$ ESA ★★

GIVEN A series circuit has $R = 40\ \Omega$, $L = 1$ H, $C = 16 \times 10^{-4}$ F and $E = 100\cos 5t$. Find the charge at any time t.

STEP 1 $\frac{1}{C} = 625$: $Q'' + 40Q' + 625Q = 100\cos 5t$.

STEP 2 AE $m^2 + 40m + 625 = 0$: $m = \frac{-40 \pm \sqrt{1600 - 2500}}{2} = -20 \pm 15i$. CF $= e^{-20t}(c_1\cos 15t + c_2 \sin 15t)$.

STEP 3 PI $= \frac{100\cos 5t}{D^2 + 40D + 625}$; $D^2 \to -25$: $\frac{100\cos 5t}{600 + 40D} = \frac{5}{2} \cdot \frac{\cos 5t}{15 + D}$.

STEP 4 Rationalise: $\frac{5}{2} \cdot \frac{(15 - D)\cos 5t}{225 - D^2} = \frac{5}{2} \cdot \frac{15\cos 5t + 5\sin 5t}{250}$.

ANSWER $\boxed{Q(t) = e^{-20t}(c_1\cos 15t + c_2 \sin 15t) + 0.15\cos 5t + 0.05\sin 5t}$

CHECK sympy dsolve and checkodesol agree ($\frac{3\cos 5t + \sin 5t}{20}$). CF roots as in the class 002 pack. Amplitude $0.158 = \frac{100}{5Z}$, $Z = 126.5\ \Omega$.

ESA Jan-24 s3 3b

W3.2 · The same ESA circuit with $Q(0) = I(0) = 0$ ESA ★

GIVEN W3.1 with the circuit uncharged and no current at $t = 0$.

STEP 1 $Q(0) = c_1 + 0.15 = 0$: $c_1 = -0.15$.

STEP 2 $Q'(t) = e^{-20t}[(-20c_1 + 15c_2)\cos 15t + (-20c_2 - 15c_1)\sin 15t] - 0.75\sin 5t + 0.25\cos 5t$.

STEP 3 $Q'(0) = -20c_1 + 15c_2 + 0.25 = 3 + 15c_2 + 0.25 = 0$: $c_2 = -\frac{13}{60} = -0.2167$.

STEP 4 $I = Q' = \frac{\cos 5t - 3\sin 5t}{4} + \frac{e^{-20t}}{12}(79\sin 15t - 3\cos 15t)$.

ANSWER $\boxed{Q(t) = -e^{-20t}\left(0.15\cos 15t + \frac{13}{60}\sin 15t\right) + 0.15\cos 5t + 0.05\sin 5t}$

CHECK sympy dsolve with the two conditions agrees; $I(0) = \frac{1}{4} - \frac{3}{12} = 0$. Figure G3.1.

ADDED FOR CLARITY ESA Jan-24 s3 3b

W3.3 · ESA: $R = 20,\ L = 1,\ C = 0.002,\ E = 12\sin 10t$ ESA ★★

GIVEN A series circuit has $R = 20\ \Omega$, $L = 1$ H, $C = 0.002$ F and $E = 12\sin 10t$. Find the charge at any time t.

STEP 1 $\frac{1}{C} = 500$: $Q'' + 20Q' + 500Q = 12\sin 10t$.

STEP 2 AE $m^2 + 20m + 500 = 0$: $m = \frac{-20 \pm \sqrt{400 - 2000}}{2} = -10 \pm 20i$. CF $= e^{-10t}(c_1\cos 20t + c_2 \sin 20t)$.

STEP 3 PI: $D^2 \to -100$: $\frac{12\sin 10t}{400 + 20D} = \frac{3}{5}\cdot\frac{\sin 10t}{20 + D} = \frac{3}{5}\cdot\frac{(20 - D)\sin 10t}{400 + 100}$.

STEP 4 $(20 - D)\sin 10t = 20\sin 10t - 10\cos 10t$, so PI $= \frac{3(20\sin 10t - 10\cos 10t)}{2500}$.

ANSWER $\boxed{Q(t) = e^{-10t}(c_1\cos 20t + c_2\sin 20t) + 0.024\sin 10t - 0.012\cos 10t}$

CHECK sympy dsolve and checkodesol agree ($\frac{6\sin 10t - 3\cos 10t}{250}$). Amplitude $0.0268 = \frac{12}{10Z}$, $Z = \sqrt{400 + 40^2} = 44.7\ \Omega$.

ESA Jan-24 s4 3b

W3.4 · The same ESA circuit with $Q(0) = I(0) = 0$ ESA ★

GIVEN W3.3 with the circuit uncharged and no current at $t = 0$.

STEP 1 $Q(0) = c_1 - 0.012 = 0$: $c_1 = 0.012$.

STEP 2 $Q'(0) = -10c_1 + 20c_2 + 0.24 = 0$: $-0.12 + 20c_2 + 0.24 = 0$, $c_2 = -0.006$.

STEP 3 $I = Q' = 0.24\cos 10t + 0.12\sin 10t - e^{-10t}(0.24\cos 20t + 0.18\sin 20t)$.

ANSWER $\boxed{Q(t) = e^{-10t}(0.012\cos 20t - 0.006\sin 20t) + 0.024\sin 10t - 0.012\cos 10t}$

CHECK sympy dsolve with the two conditions agrees; $I(0) = 0.24 - 0.24 = 0$.

ADDED FOR CLARITY ESA Jan-24 s4 3b

W3.5 · Free discharge: $L = 0.25$, $R = 250$, $C = 2 \times 10^{-6}$ (CW P36, homework) ESA ★★

GIVEN $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$ with $L = 0.25$ H, $R = 250\ \Omega$, $C = 2 \times 10^{-6}$ F, $q(0) = 0.002$ C, $\frac{dq}{dt}(0) = 0$. Find $q(t)$.

STEP 1 Divide by 0.25: $q'' + 1000q' + 2 \times 10^6 q = 0$.

STEP 2 AE: $m = \frac{-1000 \pm \sqrt{10^6 - 8 \times 10^6}}{2} = -500 \pm 500\sqrt{7}\,i$; $500\sqrt{7} = 1322.88$ (underdamped).

STEP 3 $q = e^{-500t}(c_1 \cos 1322.88t + c_2 \sin 1322.88t)$; $q(0) = c_1 = 0.002$.

STEP 4 $q'(0) = -500c_1 + 1322.88c_2 = 0$: $c_2 = \frac{0.002}{\sqrt{7}} = 0.000756$.

ANSWER $\boxed{q(t) = e^{-500t}(0.002 \cos 1322.9t + 0.000756 \sin 1322.9t)}$

CHECK sympy dsolve and checkodesol agree. The CW prints $0.0008 \sin 1323t$: the same, rounded to one figure. Figure G3.2.

CW U2 P36

Past-paper questions

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q3.b · solved in W3.1

A series circuit has $R = 40\,\Omega$, $L = 1$ H, $C = 16 \times 10^{-4}$ F and $E = 100 \cos 5t$. Find the charge at any time t.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q3.b · solved in W3.3

A series circuit has $R = 20\,\Omega$, $L = 1$ H, $C = 0.002$ F and $E = 12 \sin 10t$. Find the charge at any time t.

SHORT-NOTE RECAP

ESA s3: $Q'' + 40Q' + 625Q = 100 \cos 5t$, $Q = e^{-20t}(c_1 \cos 15t + c_2 \sin 15t) + 0.15 \cos 5t + 0.05 \sin 5t$. ESA s4: $Q'' + 20Q' + 500Q = 12 \sin 10t$, $Q = e^{-10t}(c_1 \cos 20t + c_2 \sin 20t) + 0.024 \sin 10t - 0.012 \cos 10t$. PI by $D^2 \to -\omega^2$ and rationalising. CW P36: $q = e^{-500t}(0.002 \cos 500\sqrt{7}t + 0.000756 \sin 500\sqrt{7}t)$.

4 First-order circuits: RL and RC

Without a capacitor (RL circuit) Kirchhoff's law gives $L\frac{di}{dt}+Ri=E(t)$; without an inductor (RC circuit) it gives $Ri+\frac{Q}{C}=E$, which differentiated is $R\frac{di}{dt}+\frac{i}{C}=\frac{dE}{dt}$. Both are first-order linear equations with constant coefficients, solved by the same CF + PI method with a single root: a transient $e^{-Rt/L}$ or $e^{-t/(RC)}$ plus a sinusoidal steady state.

Key terms and definitions

Time constant. $\frac{L}{R}$ (RL) or RC (RC): the time in which the transient falls to $e^{-1}\approx 37\%$.
ADDED FOR CLARITY CW U2 P34 · CW U2 P35

Formulas

E4.1 · RL circuit with a sinusoidal source ESA ★★

$$L\frac{di}{dt}+Ri=E\sin\omega t,\ i(0)=0:$$
$$i=\frac{E}{\sqrt{R^2+L^2\omega^2}}\left[\sin(\omega t-\varphi)+e^{-Rt/L}\sin\varphi\right],\ \varphi=\tan^{-1}\frac{L\omega}{R}$$

GIVES The current in an RL circuit.
APPLIES No capacitor; switch closed at $t=0$.
NOTE Transient $e^{-Rt/L}$, time constant $\frac{L}{R}$.
CW U2 P34

E4.2 · RC circuit with a cosine source ESA ★★

$$R\frac{di}{dt}+\frac{i}{C}=\frac{dE}{dt},\ E=E_0\cos pt,\ i(0)=0:$$
$$i=\frac{CpE_0}{1+p^2R^2C^2}\left[RCp\cos pt-\sin pt-pRC\,e^{-t/(RC)}\right]$$

GIVES The current in an RC circuit.
APPLIES No inductor.
NOTE From $Ri+\frac{Q}{C}=E$ differentiated; time constant RC.
CW U2 P35

Graphs and diagrams

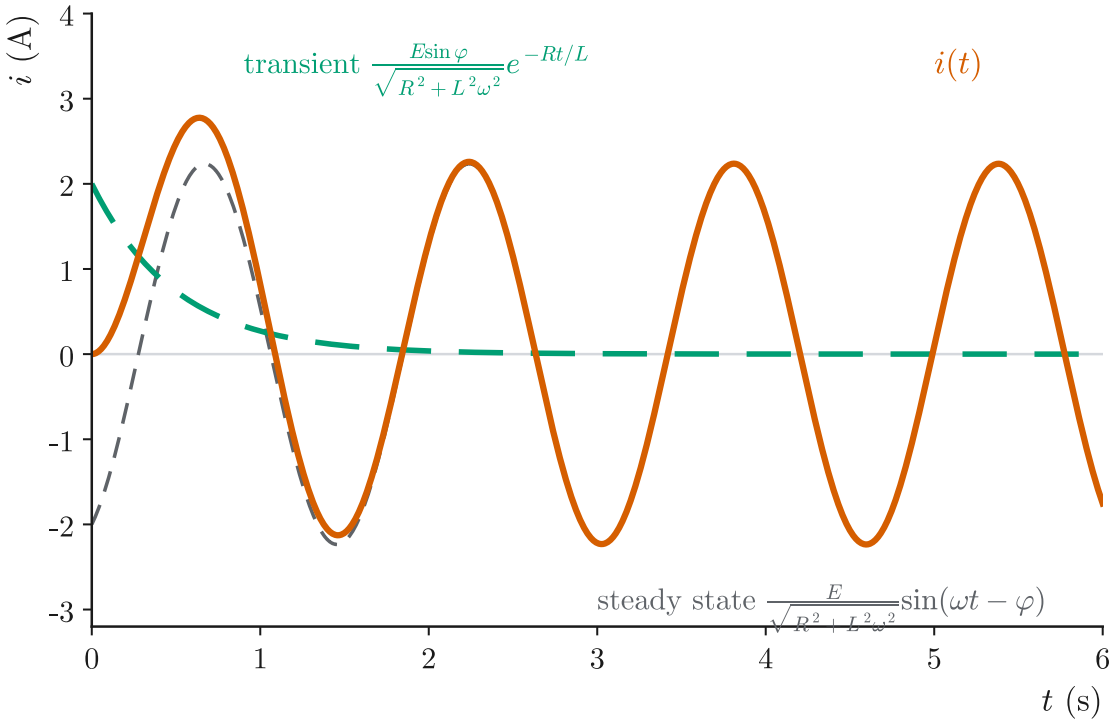
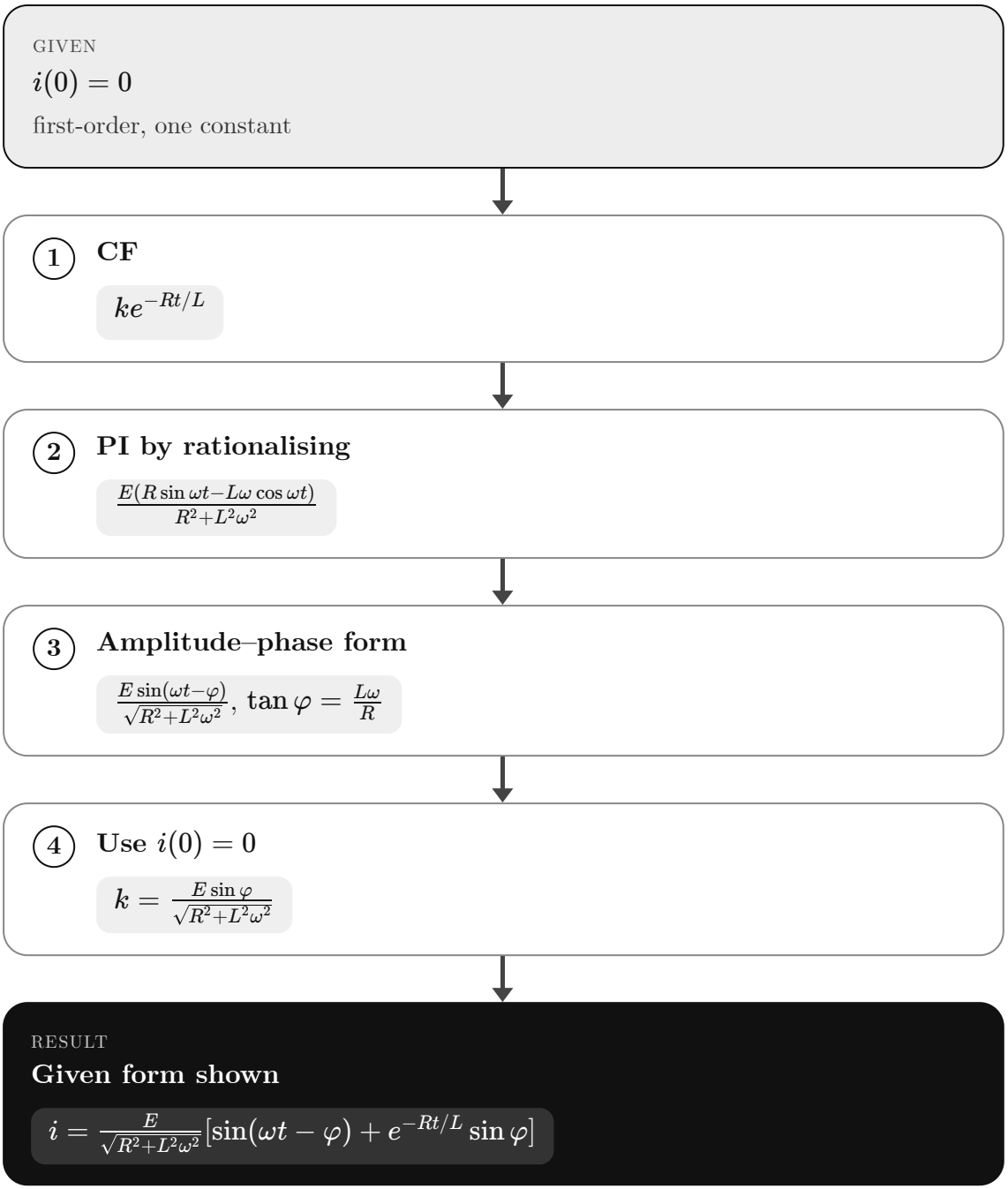


Figure G4.1. RL circuit (CW P34): current with its transient. ESA ★

ADDED FOR CLARITY CW U2 P34

How to read it. Illustrative values $E = 10$ V, $R = 2\ \Omega$, $L = 1$ H, $\omega = 4$ rad/s ($\varphi = \tan^{-1} 2$). The current (vermilion) starts at 0; the steady state $\frac{E}{\sqrt{R^2+L^2\omega^2}}\sin(\omega t-\varphi)$ (grey dashed) lags the source by φ; the difference is the transient $\frac{E\sin\varphi}{\sqrt{R^2+L^2\omega^2}}e^{-Rt/L}$ (green dashed), time constant $\frac{L}{R} = 0.5$ s.

Flowcharts



Remark. The RC circuit (W4.2) follows the same five steps with $m = -\frac{1}{RC}$.

Figure F4.1. RL circuit, $L\frac{di}{dt} + Ri = E\sin\omega t$ ESA ★★

CW U2 P34

RL, RC and LCR circuits

Circuit	Equation	Order	Transient
RL	$L\frac{di}{dt}+Ri=E(t)$	1	$e^{-Rt/L}$
RC	$R\frac{di}{dt}+\frac{i}{C}=\frac{dE}{dt}$	1	$e^{-t/(RC)}$
LCR	$L\frac{d^2Q}{dt^2}+R\frac{dQ}{dt}+\frac{Q}{C}=E(t)$	2	by R^2 vs $\frac{4L}{C}$

CW U2 P34 · CW U2 P35 · S014·6

Important points and why

1. RL: $L\frac{di}{dt}+Ri=E(t)$ is first order in i; RC: differentiate $Ri+\frac{Q}{C}=E$ to get $R\frac{di}{dt}+\frac{i}{C}=\frac{dE}{dt}$. ESA ★★ CW U2 P34 · CW U2 P35
2. A single root $m=-\frac{R}{L}$ or $-\frac{1}{RC}$ gives one constant, fixed by $i(0)=0$. ESA ★★ CW U2 P34 · CW U2 P35

Exam trap. CW P34's printed answer has $e^{-Ri/L}$ and $\sin\varphi\varphi$: read $e^{-Rt/L}\sin\varphi$. CW P35's has $e^{-1/RC}$: read $e^{-t/(RC)}$. ADDED FOR CLARITY CW U2 P34 · CW U2 P35

Worked examples

W4.1 · RL circuit: show the given form (CW P34) — ESA ★★

GIVEN $L\frac{di}{dt}+Ri=E\sin\omega t$, $i=0$ at $t=0$. Show $i=\frac{E}{\sqrt{R^2+L^2\omega^2}}[\sin(\omega t-\varphi)+e^{-Rt/L}\sin\varphi]$, $\varphi=\tan^{-1}\frac{L\omega}{R}$.

STEP 1 AE $Lm+R=0$: $m=-\frac{R}{L}$; CF $=ke^{-Rt/L}$.

STEP 2 PI $=\frac{E\sin\omega t}{R+LD}=\frac{E(R-LD)\sin\omega t}{R^2-L^2D^2}=\frac{E(R\sin\omega t-L\omega\cos\omega t)}{R^2+L^2\omega^2}$.

STEP 3 Put $R=\sqrt{R^2+L^2\omega^2}\cos\varphi$, $L\omega=\sqrt{R^2+L^2\omega^2}\sin\varphi$: PI $=\frac{E\sin(\omega t-\varphi)}{\sqrt{R^2+L^2\omega^2}}$.

STEP 4 $i(0)=k-\frac{E\sin\varphi}{\sqrt{R^2+L^2\omega^2}}=0$, so $k=\frac{E\sin\varphi}{\sqrt{R^2+L^2\omega^2}}$.

ANSWER
$$\boxed{i=\frac{E}{\sqrt{R^2+L^2\omega^2}}\left[\sin(\omega t-\varphi)+e^{-Rt/L}\sin\varphi\right],\quad \varphi=\tan^{-1}\frac{L\omega}{R}}$$

CHECK sympy dsolve minus this form simplifies to 0. The CW answer line misprints $e^{-Ri/L}\sin\varphi\varphi$. Figure G4.1.

CW U2 P34

W4.2 · RC circuit with $E = E_0 \cos pt$ (CW P35) ESA ★★

GIVEN $R\frac{di}{dt} + \frac{i}{C} = \frac{dE}{dt}$, $E = E_0 \cos pt$, $i = 0$ when $t = 0$. Solve.

STEP 1 $\frac{dE}{dt} = -E_0 p \sin pt$. AE $Rm + \frac{1}{C} = 0$: $m = -\frac{1}{RC}$; CF $= ke^{-t/(RC)}$.

STEP 2 PI $= \frac{-E_0 p \sin pt}{RD + \frac{1}{C}} = \frac{-CE_0 p \sin pt}{1 + RCD} = \frac{-CE_0 p(1 - RCD)\sin pt}{1 + R^2C^2p^2}$.

STEP 3 $(1 - RCD)\sin pt = \sin pt - RCp\cos pt$, so PI $= \frac{CpE_0(RCp\cos pt - \sin pt)}{1 + p^2R^2C^2}$.

STEP 4 $i(0) = k + \frac{CpE_0 \cdot RCp}{1 + p^2R^2C^2} = 0$: $k = -\frac{CpE_0\, pRC}{1 + p^2R^2C^2}$.

ANSWER

$$\boxed{\,i = \frac{CpE_0}{1 + p^2R^2C^2}\left[RCp\cos pt - \sin pt - pRC\, e^{-t/(RC)}\right]\,}$$

CHECK sympy dsolve minus this form simplifies to 0. The CW answer text is garbled ($I + p^2R^2C^2$, $e^{-1/RC}$).

CW U2 P35

SHORT-NOTE RECAP

RL: $Li' + Ri = E\sin\omega t$, $i(0) = 0$: $i = \frac{E}{\sqrt{R^2 + L^2\omega^2}}[\sin(\omega t - \varphi) + e^{-Rt/L}\sin\varphi]$, $\tan\varphi = \frac{L\omega}{R}$. RC: $Ri' + \frac{i}{C} = E'$, $E = E_0\cos pt$: $i = \frac{CpE_0}{1 + p^2R^2C^2}[RCp\cos pt - \sin pt - pRCe^{-t/RC}]$.

A Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
L	Inductance	Henries (H)	H	E1.1, E1.3
R	Resistance	Ohms (Ω)	Ω	E1.1, E1.3
C	Capacitance	Farads (F); not an integration constant	F	E1.1, E1.3
$E(t)$	Electromotive force	Source voltage; E, E_0 also amplitudes (CW P34–35)	V	E1.2, E4.1
Q, q	Charge	On the capacitor; slides 13–16 write q	C (coulomb)	E1.3
I, i	Current	$\frac{dQ}{dt}$	A	E1.2, E1.5
V_L, V_R, V_C	Voltage drops	Also E_L, E_R, E_C on slide 014·5	V	E1.1, E1.2
$\dot{q}$, $\ddot{q}$	Time derivatives	$\frac{dq}{dt}$, $\frac{d^2q}{dt^2}$ (slides 13–16)		E1.3
c_1, c_2	Arbitrary constants	Of Q; slide 014·16's C_1, C_2 are constants of i		E2.2
ω_d	Damped frequency	$\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$	rad/s	E2.3
ω, p	Source frequency	In $\sin \omega t$, $\cos pt$	rad/s	E2.5, E4.1, E4.2
Z	Impedance	$\sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}$	Ω	E2.5
φ	Phase angle	$\tan^{-1} \frac{L\omega}{R}$ (RL circuit)		E4.1

B Sources, coverage and gaps

Source	Path	Use
Deck 014 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.*	17 slides; content on 4–16 (every equation read from the rendered images)
Transcript T014 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.*	11.6 min; slides 4–12
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Unit 2 problems 34, 35 and 36 (all solved; pages rendered to read the answers)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Jan-24 set 3 Q3.b and set 4 Q3.b, via the shared ESA map
Video 014 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.mp4	Not opened: every slide equation is readable and no video trigger applies

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). LCR circuits: 2 distinct questions, 2 occurrences, both 6 marks and both in the Jan 2024 sets: set 3 Q3.b ($R = 40\,\Omega$, $L = 1$ H, $C = 16 \times 10^{-4}$ F, $E = 100\cos 5t$) and set 4 Q3.b ($R = 20\,\Omega$, $L = 1$ H, $C = 0.002$ F, $E = 12\sin 10t$), each "find the charge at any time t" with no initial conditions (★★). CW P34–P36 add the RL and RC circuits and a free LCR discharge (★★). Never asked: the derivation of the model as a question on its own, the current equation, the damping classification and the resonance/impedance view (★).

Known gaps, conflicts and corrections

1. Every slide answer in this class is right: slides 014·8–12 ($Q = e^{-2t}(-\cos t - 2\sin t) + 1$, $I = 5e^{-2t}\sin t$) and slides 014·13–16 ($C_1 = -0.0423597$, $C_2 = 0.526168$, $A = -0.00128331$, $B = -0.00366164$) were recomputed with sympy dsolve and checkodesol.
2. Slide 014·16 finds C_1, C_2 "using initial conditions" but states only $i(0) = 0$; the second condition is $i'(0) = 0$, which follows from the equation at $t = 0$: $\ddot{q}(0) = 1550\sin 0 - 1000\dot{q}(0) - 10000q(0) = 0$ (added for clarity, W2.2).
3. Slide 014·11 writes $e^{-2.0}$, meaning $e^{-2\cdot 0} = 1$. Slide 014·5 spells Kirchhoff as "Kirchoff". Slide 014·12's equation (3) already has $c_1 = -1$ substituted.
4. CW P34 prints its answer as $\frac{E}{\sqrt{R^2+L^2\omega^2}}[\sin(\omega t - \varphi) + e^{-Ri/L}\sin\varphi\varphi]$; the exponent is $-\frac{R}{L}t$ and the factor is $\sin\varphi$ (as in the question). "If $i = 0$" means $i = 0$ at $t = 0$. The stated form is correct (sympy).

5. CW P35's printed answer is garbled ($I + p^2R^2C^2$, $e^{-1/RC}$). Recomputed: $i = \frac{CpE_0}{1+p^2R^2C^2}\left[RCp\cos pt - \sin pt - pRC\,e^{-t/(RC)}\right]$ (sympy).
6. CW P36 prints $q = e^{-500t}(0.002\cos 1323t + 0.0008\sin 1323t)$. Exactly, $\omega_d = 500\sqrt{7} = 1322.88$ and the sine coefficient is $\frac{0.002}{\sqrt{7}} = 0.000756$; the printed 0.0008 is that value rounded to one figure (5.8 % high), not an independent slip.
7. The damping classification by R^2 against $\frac{4L}{C}$, the current equation $LI'' + RI' + \frac{I}{C} = E'(t)$, the steady-state amplitude through the impedance, and the zero-initial-condition versions of both ESA questions are added for clarity.
8. Both ESA questions give no initial conditions, so the expected answer is the general solution with two arbitrary constants; W3.2 and W3.4 also give the charge for $Q(0) = I(0) = 0$ (added for clarity). Their CF roots ($-20 \pm 15i$, $-10 \pm 20i$) agree with the class 002 pack.
9. The lecture says the capacitance is "0.15" and $E = 10$ "ohms" (T014 00:04:29, 00:05:17); the slide gives $C = 0.1$ farad and $E = 10$ volts, and the working uses those values.
10. Transcript recognition errors were corrected silently: "LCI circuit", "L Cr", "NCR side" → LCR circuit; "K chop's", "KCO's" → Kirchhoff's; "poses" → ohms; "fat" → farads; "holes" → coulombs; "IFP", "IFD", "If P" → $I(t)$; "ESP", "EFT", "CFP" → $E(t)$; "q of p", "Q of D" → $Q(t)$; "DIB" → $\frac{dI}{dt}$; "voltage log" → voltage drop; "electromotte" → electromotive; "phi by d square" → $\frac{5}{D^2+4D+5}$; "2 degrees" → 2 henry; "electric circulates" → electric circuits.
11. Not in this class: spring–mass (SHM) applications are in the class 015 pack.

Coverage

Source	Citations	Slides cited (slide × citations)
S014	94	4×6, 5×8, 6×8, 7×6, 8×5, 9×15, 10×7, 11×7, 12×9, 13×2, 14×2, 15×13, 16×6
CW	22	
ESA	29	
T014	31	

Slides 1–3 and 17 are title, class-content and closing slides; slides 4–12 are taught in the transcript (T014, 11.6 min, ending with slide 12); slides 13–16 (the $155\sin 377t$ example, typeset differently) have no lecture and are cited by slide alone. Bucket numbers are not contiguous: the deck's title slide says SESSION 10, and the bucket numbers 007, 008, 010, 012 and 013 do not exist (009 = session 7, 011 = sessions 8 and 9, 014 = 10, 015 = 11).